

[1] Probability Theory

'26.06.29 (Mon)

①

Statistical regularity $\leadsto$ probability model.

[1.1]. Terminologies: $\left\{ \begin{array}{l} \bullet \text{ experiment} \\ \bullet \text{ sample space (e.g. } S) \\ \bullet \text{ event (e.g., } A, B) \end{array} \right.$

(e.g.)

$$S = \{H, T\}$$

$$S = \{HH, HT, TH, TT\}, \quad S = \{T, HT, HHT, HHHT, \dots\}, \quad S = \{x \mid 0 \leq x < \infty\}.$$

Def 1.1

Intersection: $A \cap B$.

Union: $A \cup B$

Def. 1.2 (mutually exclusive)

$$A \cap B = \emptyset$$

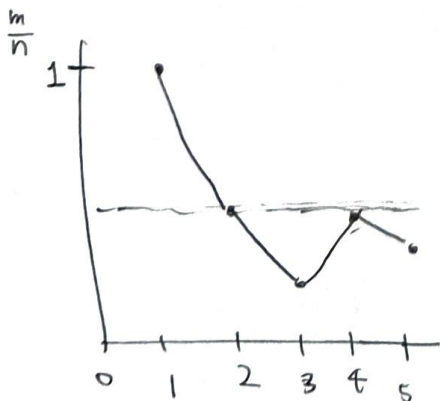
Def 1.3 (complement)

For $A \subset S$, the set $\{e \mid e \notin A \wedge e \in S\} =: A^c$ is 'complement of A '.

[1.2]. Classical def. of prob. $\leadsto \frac{M}{N} \left(\frac{\# \text{event}}{\# \text{sample}} \right)$.

$\hookrightarrow$ relative frequency

$$\leadsto P(A) := \lim_{n \rightarrow \infty} \frac{m}{n}$$



$n \rightarrow \infty$
 $\Rightarrow$



②.

Axiom of Probability (Kolmogorov).

(1) For arbitrary event A , $P(A) \geq 0$.

(2) $P(S) = 1$.

(3) Let $A_1, A_2, \dots \subset S$. For any $i \neq j$, if $A_i \cap A_j = \emptyset$, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Thm 1.1

(1) $P(A^c) = 1 - P(A)$.

(2) $P(\emptyset) = 0$

(3) $P(A) \leq P(B)$ if $A \subset B$.

(4) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

pf) (1) $1 = P(S) = P(A \cup A^c) = P(A) + P(A^c)$ ($\because$ Axiom (3).)

(2) $P(\emptyset) = P(S^c) = 1 - P(S) = 0$. ($\because$ Axiom (2) and (1)).

(3). $B = A \cup (B \cap A^c) \leadsto A \cap (B \cap A^c) = \emptyset$.

$\Rightarrow P(B) = P(A) + P(B \cap A^c) \geq P(A)$ ($\because$ Axiom (1) and (3)).

(4) $P(A) = P(A \cap B^c) + P(A \cap B)$

$P(B) = P(B \cap A^c) + P(B \cap A)$

□

Example 1.11.

$$P(A \cup B \cup C) = P((A \cup B) \cup C)$$

$$= P(A \cup B) + P(C) - P((A \cup B) \cap C)$$

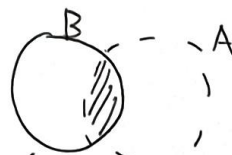
$$= \dots$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C).$$

1-3

Def 1.4 (Conditional probability)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

sample space $\leftarrow S$.

($\because$ If we know the event B happened, then we cannot observe B^c .)

PropertyFor $A_1 \cap A_2 = \emptyset$,

$$\begin{aligned} P(A_1 \cup A_2 | B) &= \frac{P[(A_1 \cup A_2) \cap B]}{P(B)} \\ &= \frac{P((A_1 \cap B) \cup (A_2 \cap B))}{P(B)} = \frac{P(A_1 \cap B) + P(A_2 \cap B)}{P(B)} \\ &= P(A_1 | B) + P(A_2 | B). \end{aligned}$$

$$\rightarrow P(A^c | B) = 1 - P(A | B),$$

$$P(\emptyset | B) = 0, \quad P(A_1 | B) \leq P(A_2 | B) \text{ if } A_1 \subset A_2,$$

$$P(A_1 \cup A_2 | B) = P(A_1 | B) + P(A_2 | B) - P(A_1 \cap A_2 | B). \quad \text{hold.}$$

Ex. 1.14

$$P(W_1 \cap W_2) = P(W_1) P(W_2 | W_1).$$

Ex. 1.15

$$\begin{aligned} &P(W_1 \cap W_2 \cap W_3 \cap R) \\ &= P(W_1) \cdot P(W_2 | W_1) \cdot P(W_3 | W_1 \cap W_2) \cdot P(R | W_1 \cap W_2 \cap W_3) \end{aligned}$$

④.

Thm 1.2 (Law of probability)

For the events $B_1, B_2, \dots, B_k$ s.t. $B_i \cap B_j = \emptyset, i \neq j$,
let $\bigcup_{i=1}^k B_i = S$. Then for any event A ,

$$P(A) = \sum_{i=1}^k P(B_i) P(A|B_i)$$

Thm 1.3 (Bayes Thm.)

Let $B_i \cap B_j = \emptyset, i \neq j$. and $\bigcup_{i=1}^k B_i = S$.

$$P(B_j|A) = \frac{P(B_j) P(A|B_j)}{\sum_{i=1}^k P(B_i) P(A|B_i)}$$

(pf) $P(B_j|A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(B_j) P(A|B_j)}{P(A)}$

by Thm 1.2, $P(A) = \sum_{i=1}^k P(B_i) P(A|B_i)$ $\square$

Def. 1.5 (Independent).

$$P(A \cap B) = P(A) P(B).$$

Def. 1.6 (Mutually independent)

For n events, $A_1, A_2, \dots, A_n$ and k index sets $i_1, i_2, \dots, i_k$
($k=1, \dots, n$),

$$P(A_{i_1} \cap A_{i_2} \dots \cap A_{i_k}) = P(A_{i_1}) P(A_{i_2}) \dots P(A_{i_k})$$

$\Rightarrow A_1, A_2, \dots, A_n$: "mutually independent".

1.4 number of cases.

• Permutation: $nPr = n(n-1)(n-2) \dots (n-r+1)$

$$= \frac{n!}{(n-r)!}$$

• Combination: $nCr = \frac{nPr}{r!} = \frac{n!}{r!(n-r)!} = \binom{n}{r}$.

ex. 1.23

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

$\underbrace{\qquad\qquad\qquad}_{\text{binomial coefficient.}}$

Take $a=b=1 \Rightarrow 2^n = \sum_{k=0}^n \binom{n}{k}$.

For k -kind, s.t. $n = r_1 + r_2 + \dots + r_k$ and $r_i (i=1, 2, \dots, k)$,
the multinomial coefficient is:

$$\binom{n}{r_1, r_2, \dots, r_k} = \frac{n!}{r_1! r_2! \dots r_k!}$$

Ex. 1.24 (Multinomial Thm.)

$$(a_1 + a_2 + \dots + a_k)^n = \sum_{r_1 + \dots + r_k = n} \binom{n}{r_1, r_2, \dots, r_k} a_1^{r_1} a_2^{r_2} \dots a_k^{r_k}$$

Ex. 1.29

- Sampling with replacement. (considers the order)
- " without " (it doesn't matter)

⑥ Chap 1 Exercise.

#2. $k > 3, A_1, \dots, A_k,$

$$P\left(\bigcup_{i=1}^k A_i\right) = \sum_{i=1}^k P(A_i) - \sum_{1 \leq i < j \leq k} P(A_i \cap A_j) + \dots + (-1)^{k-1} P\left(\bigcap_{i=1}^k A_i\right)$$

(pf) Inclusion - Exclusion principle.

Use mathematical induction.

1. Base Step: $k=2$

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).$$

2. Inductive Hypothesis: $k=m$ ($m \geq 2$).

$$\text{Let } P\left(\bigcup_{i=1}^m A_i\right) = \sum_{i=1}^m P(A_i) - \sum_{1 \leq i < j \leq m} P(A_i \cap A_j) + \dots + (-1)^{m-1} P\left(\bigcap_{i=1}^m A_i\right).$$

3. Inductive Step.

$$\text{Note that } A_{m+1} \cup \underbrace{\left(\bigcup_{i=1}^m A_i\right)}_{=: B} = \bigcup_{i=1}^{m+1} A_i$$

$$P(B \cup A_{m+1}) = P(B) + P(A_{m+1}) - P(B \cap A_{m+1}) \quad \dots (*)$$

$$① \quad P(B) = \sum_{i=1}^m P(A_i) - \sum_{1 \leq i < j \leq m} P(A_i \cap A_j) + \dots + (-1)^{m-1} P\left(\bigcap_{i=1}^m A_i\right) \quad \text{by inductive hypothesis.}$$

$$② \quad P(B \cap A_{m+1}) = \left(\bigcup_{i=1}^m A_i\right) \cap A_{m+1} = \bigcup_{i=1}^m (A_i \cap A_{m+1}), \quad \text{apply the inductive hypothesis again.}$$

$$P(B \cap A_{m+1}) = \sum_{i=1}^m P(A_i \cap A_{m+1}) - \sum_{1 \leq i < j \leq m} P(A_i \cap A_j \cap A_{m+1}) + \dots + (-1)^{m-1} P\left(\bigcap_{i=1}^m A_i\right)$$

$$(*) = \dots = (\text{above expression holds.}) \quad \square$$

#3. (Boole's Ineq.)

$$\text{Prove } P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n).$$

(pf.)

Construct a new sequence of sets $B_1, B_2, B_3, \dots$

$$B_1 = A_1$$

$$B_2 = A_2 \setminus A_1$$

$$B_3 = A_3 \setminus (A_1 \cup A_2)$$

$$\vdots$$

$$B_n = A_n \setminus \left(\bigcup_{j=1}^{n-1} A_j\right)$$

Then sets B_n are mutually disjoint. ($B_i \cap B_j = \emptyset$ for $i \neq j$)and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$. Note that each new set is a subset oforiginal set. i.e., $B_n \subseteq A_n \Rightarrow P(B_n) \leq P(A_n)$ by monotonicity.

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = P\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} P(B_n) \leq \sum_{n=1}^{\infty} P(A_n)$$

$$\therefore P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n) \quad \square$$

#4. (Bonferroni's Inequality)

$$\text{Prove that } P\left(\bigcap_{n=1}^k A_n\right) \geq 1 - \sum_{n=1}^k P(A_n^c)$$

(pf.)

$$P\left(\bigcap_{n=1}^k A_n\right) = 1 - P\left(\left(\bigcap_{n=1}^k A_n\right)^c\right) = 1 - P\left(\bigcup_{n=1}^k A_n^c\right) \geq 1 - \sum_{n=1}^k P(A_n^c)$$

$$(\because \text{Boole's Ineq.}) \quad \square$$

8.

#17. For $n > r > 0$, prove that

$$(1) \binom{n}{r} = \binom{n}{n-r}$$

$$(2) \binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$$

(pf). (1) $\binom{n}{r} = \frac{n!}{(n-r)! r!} = \frac{n!}{r! (n-r)!} = \binom{n}{n-r}$

$$\begin{aligned} (2) \binom{n-1}{r-1} + \binom{n-1}{r} &= \frac{(n-1)!}{(n-1-(r-1))! (r-1)!} + \frac{(n-1)!}{(n-1-r)! r!} \\ &= \frac{(n-1)!}{(n-r)! (r-1)!} + \frac{(n-1)!}{(n-1-r)! r!} \\ &= \frac{(n-1)! r + (n-1)! (n-r)}{(n-r)! r!} = \frac{(n-1)! (r + (n-r))}{(n-r)! r!} \\ &= \frac{(n-1)! n}{(n-r)! r!} = \frac{n!}{(n-r)! r!} \quad \square \end{aligned}$$